👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3C

Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratio reasoning to convert measurements between the customary and metric systems.

Language Objective: I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Conversion factor · **Customary system** · **Metric system** · **Approximately** · **Ratio table** · **Reasonableness** · **Multi-step problem**

Level 3: try the blanks from memory first, then check the bank.

1 The ratio that trades one unit for another is a _____ _____.

2. Inches, pounds and gallons belong to the _____ system.

3. Centimeters, grams and liters belong to the _____ system, built on tens.

4. Close to but not exactly equal is _____ — written with ≈.

5. A table whose every column holds the same comparison is a _____ _____.

6. Checking that an answer's size makes sense is checking its _____.

7. A problem that needs two or more operations in order is a _____.

Write About the Math

Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?
¿100 kilómetros por hora es más rápido o más lento que 100 millas por hora, y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Conversion factor** — *Factor de conversión*
- ☐ **Customary system** — *Sistema usual*
- ☐ **Metric system** — *Sistema métrico*
- ☐ **Approximately** — *Aproximadamente*
- ☐ **Ratio table** — *Tabla de razones*

Model: *A mile is longer than a kilometer (1 km ≈ 0.6 mi), so 5 miles is the longer race even though the numbers match.* — Students reason about UNIT SIZE before touching arithmetic — same number, different units, different lengths.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The conversion factor is 1 km ≈ ___ mi. Scaling that up, 100 kilometers ≈ ___ miles, so the sign means about ___ miles per hour. That is ___ than 100 miles per hour, because in one hour you would travel ___ distance.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Conversion factor
2. **Notes 2:** Customary system
3. **Notes 3:** Metric system
4. **Notes 4:** Approximately
5. **Notes 5:** Ratio table
6. **Notes 6:** Reasonableness
7. **Notes 7:** Multi-step problem

8. **Watch for this mistake:** *Compare measurements only after converting to a common unit. Also check whether the conversion factor is exact or rounded: 1 inch = 2.54 cm and 1 international mile = 1.609344 km are exact; 1 mile ≈ 1.609 km is rounded. Use ≈ when the factor or the result has been rounded.*
9. **Try It 1:** *— 1 inch = 2.54 centimeters exactly. The other displayed factors are rounded: 1 mile ≈ 1.609 km, 1 pound ≈ 0.454 kg, and 1 United States liquid quart ≈ 0.946 L.*
10. **Try It 2:** A. The conversion factors are rounded, so results are approximate *— One international mile equals exactly 1.609344 kilometers. The shorter factor 1.609 is rounded, so a result calculated with 1.609 is approximate. Exact factors, such as 1 inch = 2.54 cm, do not require ≈.*